

AEQUITAS

A Human-Centric Decentralized Monetary System Proof of Humanity Blockchain with Algorithmic Monetary Justice

Whitepaper v0.9 | June 2026 | github.com/hanoi96international-gif/Aequitas

*"Money exists because people exist. Nothing more,
nothing less."*

Abstract. Aequitas is the first decentralized monetary system that ties money supply directly to verified human existence. Every verified human receives 1,000 AEQ upon registration — unconditionally, equally, and permanently. The total supply is capped at verified humans x 1,000 AEQ. A novel Proof of Humanity consensus mechanism ensures equal validator rights for all. Algorithmic monetary policy driven exclusively by on-chain data adjusts inflation between 0% and 1.5% annually. A dynamic wealth cap with waterfall redistribution prevents extreme concentration. The system is fully deployed on Ethereum Sepolia with real Groth16 ZKP biometric verification, a 24/7 proof server on Railway, and an automated keeper bot — all operational today.

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1. The Problem with Money

The global monetary system has three fundamental flaws:

1.1 Arbitrary Money Supply. Central banks can expand the money supply at will. New money flows first to financial institutions — the Cantillon Effect — continuously transferring wealth from the many to the few.

1.2 Structural Exclusion. 1.4 billion adults worldwide have no access to banking services. The system was designed for those who already have capital.

1.3 Early Adopter Inequality. Existing cryptocurrencies replicate inequalities. Bitcoin miners and early buyers hold disproportionate wealth with no mechanism for equal participation.

Aequitas addresses all three problems simultaneously through a single, coherent design.

2. Core Principles

Human-Backed Supply. Total AEQ supply = verified humans x 1,000. Money cannot be created without a verified human.

Equal Start. Every verified human receives exactly 1,000 AEQ. No exceptions. No early adopter advantage.

Algorithmic Justice. Monetary policy determined entirely by on-chain data. No human can override it.

Decentralized Identity. Biometric ZKP verification. Zero biometric data leaves the device.

No Banks. No State.. Immutable smart contracts. No entity can freeze accounts or modify balances.

3. Proof of Humanity — Biometric Identity

The central challenge is Sybil resistance: ensuring each human registers only once.

3.1 Current Implementation

The Android app (Aequitas Human Proof) uses the device Hardware Secure Element (fingerprint or Face ID) with a deterministic payload. The biometric signature is generated locally and never transmitted.

```
payload = 'aequitas_identity_v4_permanent' → always same signature for same finger
```

3.2 Proof Server Architecture

A 24/7 Node.js server on Railway receives the biometric hash and generates a real Groth16 ZKP proof. The server performs triple Sybil protection:

- Check 1: Is the wallet already registered on-chain? (isHuman mapping)
- Check 2: Is the biometric hash already in memory? (cross-wallet protection)
- Check 3: Is the commitment already used on-chain? (usedCommitments mapping)

```
commitment = ZKP( biometricHash x walletAddress + deviceSalt )
```

3.3 Phase 1: PPG Cardiac Biometrics

The next evolution introduces Photoplethysmography (PPG) as a biometric factor. Each person's cardiac rhythm is unique and cannot be photographed or forged. A MAX30102 PPG sensor (\$3-5) provides medical-grade accuracy. This is believed to be the world's first application of PPG for decentralized identity.

Triple Sybil protection: same wallet → blocked on-chain. Same biometric → blocked in memory. Same commitment → blocked on-chain.

4. Economic Model

4.1 Initial Grant. Every verified human receives 1,000 AEQ upon registration. Unconditional and permanent.

4.2 Supply Cap. Maximum Supply = Verified Humans x 1,000 AEQ. Immutable.

Maximum Supply = Verified Humans x 1,000 AEQ

4.3 Transaction Fee Distribution (0.1% per transaction)

Recipient	Share	Purpose
Validators	40%	Network security and block production
Liquidity Providers	30%	DEX liquidity (verified humans only)
UBI Pool	20%	Distributed equally to all humans monthly
Treasury	10%	Development, audits, ecosystem grants

5. Monetary Policy

Inflation is calculated entirely from on-chain data (0%–1.5% annually):

`inflationRate = f(velocity, activeRatio, gini, growthRate) – all on-chain, no oracle`

All inflation distributed equally to every verified human. No deflation — the wealth cap mechanism replaces it.

6. Wealth Distribution Mechanism

"You can accumulate as much as you want. Beyond a threshold, the excess flows to everyone else. Automatically. Immediately."

Phase	Registrations	Maximum Balance	Status
0	0 – 100	No cap	Testing phase
1	100 – 1,000	20x fairShare	Observation
2	1,000 – 10,000	10x fairShare	Soft redistribution
3	10,000 – 100,000	5x fairShare	Full mechanism
4	100,000+	3–5x (Gini-dynamic)	Fully autonomous

OECD research: fairest societies have 5:1 wealth ratio. Aequitas targets this as steady-state.

7. System Architecture

Component	Technology	Status
AequitasV5 Contract	Solidity, Ethereum Sepolia	✓ Live
Bio Verifier (Groth16)	snarkjs/circom, on-chain	✓ Live
Android App (Aequitas Human Proof)	React Native 0.74, Hardware Secure Element	✓ Live
Proof Server	Node.js, Railway.app	✓ 24/7 Online
Keeper Bot	Node.js, Railway.app	✓ Hourly cycles
Web DApp	HTML/JS, GitHub Pages	✓ Live
Cosmos SDK Chain	Go, Ignite CLI	■ Phase 1
BlockDAG Consensus	Custom implementation	■ Phase 2

8. Proof Server & Sybil Protection

The Aequitas Proof Server runs 24/7 on Railway and handles real Groth16 ZKP generation:

```
POST /prove {bio, salt, wallet} → {pA, pB, pC, pubSignals}
```

The server implements triple Sybil protection without a central database:

1. On-chain check: `contract.isHuman(wallet)` — $O(1)$ blockchain lookup
2. In-memory check: Set of bio hashes — blocks cross-wallet reuse
3. On-chain check: `contract.usedCommitments(commitment)` — cryptographic finality

```
Server: https://aequitas-proof-server-production.up.railway.app
```

9. Consensus Mechanism: Proof of Humanity

1 verified human = 1 validator = 1 equal vote

No capital requirement. No hardware requirement. Every verified human has equal validator rights.

Mechanism	Who Decides	Democratic?
Proof of Work	Those with most hardware	No
Proof of Stake	Those with most capital	No
Proof of Humanity (Aequitas)	Every verified human equally	Yes

10. Validator Rewards

Validators rewarded exclusively from transaction fees — no new AEQ created. Each human capped at 10 blocks/day to prevent hardware advantages.

11. Lending Protocol

Peer-to-peer credit for verified humans. Max loan = 50% of balance. No compound interest. No credit score required. Liquidation at 120% collateral.

12. Governance

Dual-token model: AEQ (currency, every human) and Governance Token (voting, investors). Core parameters immutable — only adjustable parameters subject to governance.

13. Roadmap

Phase 0 — Complete ✓

- Whitepaper v0.1–0.9
- AequitasV4 + V5 contracts deployed on Ethereum Sepolia
- Real Groth16 ZKP biometric verification on-chain
- Android app (Aequitas Human Proof) with Hardware Secure Element
- Proof Server on Railway — real ZKP generation 24/7
- Keeper Bot on Railway — automatic monetary cycles hourly
- Triple Sybil protection implemented and tested
- Live Web DApp on GitHub Pages

Phase 1 — In Progress

- MAX30102 PPG sensor integration (hardware ordered)
- Multi-factor biometric: PPG + fingerprint
- Cosmos SDK chain scaffold
- Ethereum Foundation Grant application
- iOS application

Phase 2 — Testnet

- Public Aequitas testnet launch
- BlockDAG architecture implementation
- Lending protocol beta
- Independent security audit

Phase 3 — Mainnet

- Mainnet launch
- DEX liquidity provision
- Cross-chain bridges
- Governance token distribution
- Global registration campaign

14. Technical Specifications

Parameter	Value
Token Name	Aequitas (AEQ)
Decimals	18
Initial Grant	1,000 AEQ per verified human
Supply Cap	Verified Humans x 1,000 AEQ
Transaction Fee	0.1%
Max Annual Inflation	1.5% (algorithmic, 0%–1.5%)
Wealth Cap Phase 3	5x fairShare (dynamic)
Consensus Phase 2	Proof of Humanity (BlockDAG)
Current Network	Ethereum Sepolia (Testnet)
Contract V5	0x4f147d5B3388AF07993CC4fC548502A78Af0B8b5
Contract V4	0x2B5ACedF2c41c70d51A2cbAd927b8940EE725DA7
Bio Verifier	0x39Ac1431C94F6391B92d39615aB56B888Bbf2389
ZKP System	Groth16 (snarkjs / circom 2.x)
Proof Server	https://aequitas-proof-server-production.up.railway.app
Android App	Aequitas Human Proof (React Native 0.74)
Biometric Current	Hardware Secure Element (fingerprint/Face ID)
Biometric Phase 1	PPG (MAX30102) + Hardware Secure Element
TPS Phase 1	~10,000 (Cosmos SDK)
TPS Phase 2	100,000+ (BlockDAG)

15. Open Research Questions

Device-Independent Biometrics. Full device independence requires fuzzy commitment schemes for biometric templates. The current server-side Sybil protection partially addresses this through in-memory bio-hash tracking.

PPG Stability at Scale. Calibrating fuzzy commitment tolerances for global populations requires large-scale empirical studies across diverse demographics.

BlockDAG Proof of Humanity Consensus. One-human-one-vote consensus on a BlockDAG is a novel distributed systems problem with no prior precedent.

Optimal Monetary Parameters. The algorithmic inflation formula requires empirical validation through simulations and real-world pilots.

Long-term Sybil Resistance. The in-memory bio-hash cache is lost on server restart. A persistent, privacy-preserving biometric registry is an open research problem.

16. Conclusion

The global monetary system has failed billions of people — not through accident, but through design.

Aequitas is the first attempt to invert this relationship fundamentally. The system is not theoretical — it is operational today:

- Real Groth16 ZKP biometric verification on Ethereum Sepolia
- 24/7 proof server generating real cryptographic proofs
- Automated keeper bot running monetary cycles hourly
- Triple Sybil protection — tested and verified
- Android app (Aequitas Human Proof) — available today

"The first monetary system in history that belongs to no state, no bank, no corporation. Only to humanity."

GitHub: github.com/hanoi96international-gif/Aequitas

DApp: hanoi96international-gif.github.io/Aequitas/aequitas-dapp.html

Proof Server: aequitas-proof-server-production.up.railway.app

Contract V5: 0x4f147d5B3388AF07993CC4fC548502A78Af0B8b5 (Sepolia)